

List of experiments:

- 1) Introduction to Matlab Software and Work Environment.
Introduction to Matlab basic Mathematical, trigonometric, Array and Matrix operations.
- 2) To Program various logical and Iterative loops in Matlab scripts
- 3) To Program Gauss elimination & Gauss-Jordan elimination method for solving linear equations and its validation with Matlab functions
- 4) To Program Jacobi Method & Gauss Seidel method for solving linear equations and its validation with Matlab functions
- 5) To Program Diagonalization methods for solving Matrices and Eigen values
- 6) To Program Power method for solving Matrices and Eigen values and its validation with Matlab functions
- 7) To Program Lagrange Polynomial interpolation method in Matlab scripts
- 8) To Program Bisection method & Regula Falsi method of solving nonlinear equation and its validation with Matlab functions
- 9) To Program Secant method and Newton-Raphson method of solving non-linear equation and its validation with Matlab functions
- 10) To Program Euler's method of solving ordinary differential equations and its validation with Matlab functions
- 11) To Program Runge-Kutta method of solving ordinary differential equations and its validation with Matlab functions.